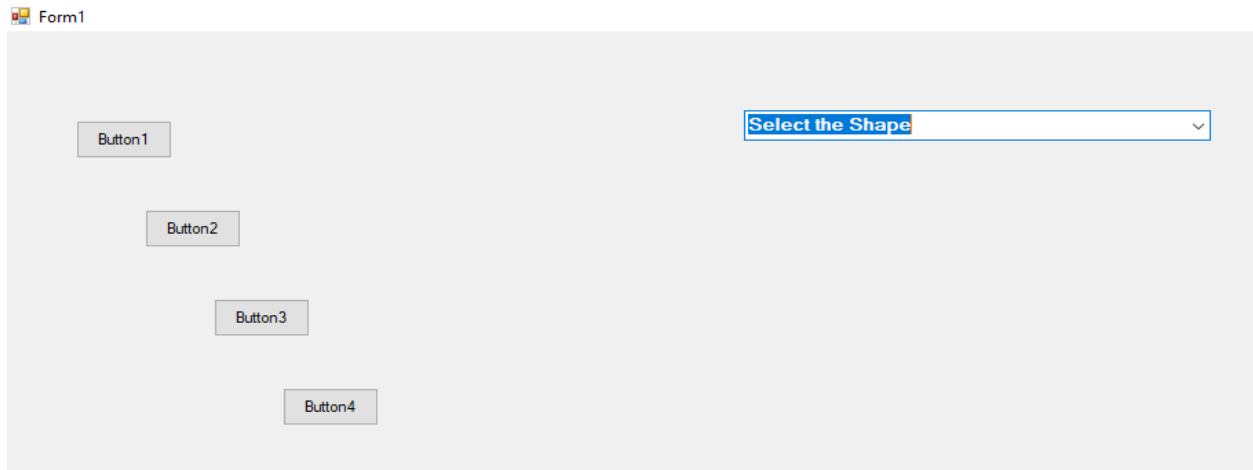


Task 2:



```
1 reference
private void Form1_Load(object sender, EventArgs e)
{
    Button[] b = new Button[5];
    for (int i = 1; i <= b.Length - 1; i++)
    {
        b[i] = new Button();
        b[i].Text = "Button" + i;
        b[i].Size = new Size(70, 30);
        b[i].Location = new Point(i * 70, i * 100);
        b[i].Left = 50 * i;
        b[i].Top = i * 70;
        b[i].Click += new System.EventHandler(button1_Click);
        this.Controls.Add(b[i]);
    }
}

1 reference
private void button1_Click(object sender, EventArgs e)
{
}
}
```

Task 3:

```
private void Form1_Load(object sender, EventArgs e)
{
    pictureBox1.SizeMode = PictureBoxSizeMode.StretchImage;

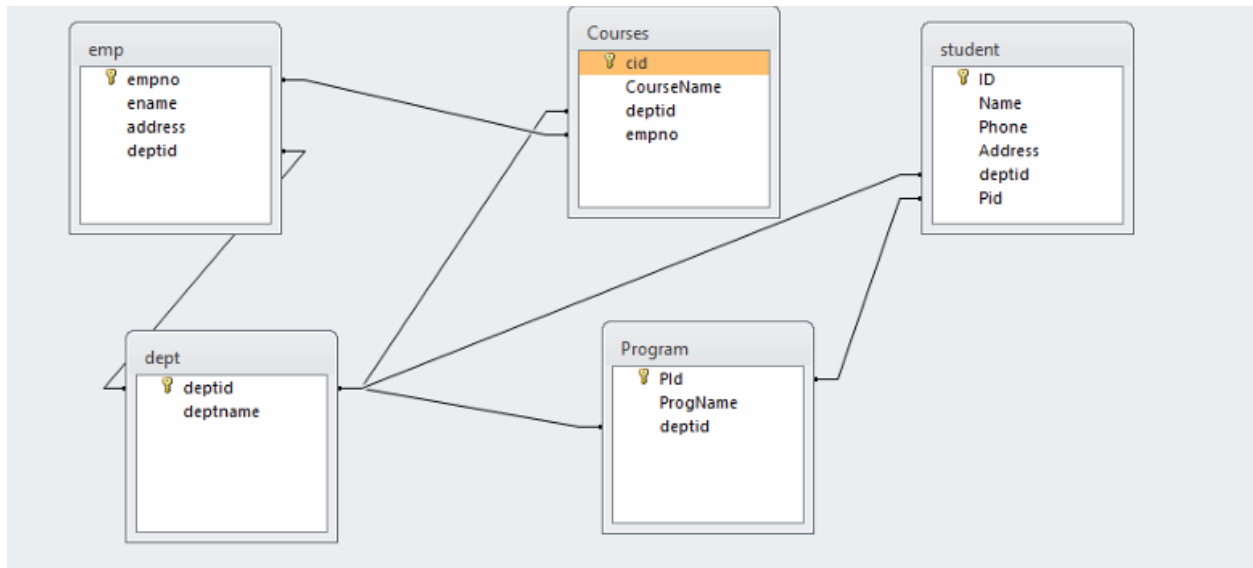
    string[] allowedExtensions = new[] { ".jpg", ".jpeg", ".png", ".bmp", ".gif" };

    imagePaths = Directory.GetFiles(imageFolder, "**.*",
        SearchOptions.CaseInsensitive)
        .Where(file => allowedExtensions.Contains(Path.GetExtension(file).ToLower()))
        .OrderBy(file => file)
        .ToList();

    if (imagePaths.Count > 0)
    {
        ShowImage(imageIndex);
        timer1.Start();
    }
}

private void timer1_Tick(object sender, EventArgs e)
{
    imageIndex++;
    if (imageIndex >= imagePaths.Count)
        imageIndex = 0;
    ShowImage(imageIndex);
}
```

Task 4:



Form1

Student ID:

Student Name:

Student Phone:

Student Address:

Student Deptid:

Display

Form4

Form2

Form3

Join

First

Previous

Next

Last

Display in textBoxes

Insert

Update

Delete

Form2

Seach For student Info

Submit Delete

The screenshot shows a Windows Form window titled 'Form2'. The form has a light gray background. At the top center, the text 'Seach For student Info' is displayed in a black, italicized font. Below this text is a white rectangular search input field with a small downward arrow on its right side. Underneath the search field are two buttons: 'Submit' on the left and 'Delete' on the right. Below these buttons is a large, empty rectangular area with a gray background, intended for displaying search results.

Properties

Form1 System.Windows.Forms.Form

(DataBindings)

(Advanced)

Tag (None)

Text nk.AccountNo

DataSet11

- bank
 - AccountNo
 - AccountType
 - ATMReq
 - Balance

The screenshot shows the 'Properties' window in Visual Studio. The 'Form1' is selected, and its properties are listed. The 'Text' property is set to 'nk.AccountNo'. The 'DataBindings' section is expanded, showing a tree view of the data source. The tree shows 'DataSet11' containing a 'bank' table, which has four fields: 'AccountNo', 'AccountType', 'ATMReq', and 'Balance'. The 'AccountNo' field is highlighted.

Form1

Account Number :

Account Holder Name :

Select Branch

Balance :

Account Type

☐ Current

☐ Saving

☐ ATM

Next Previous Add Update Exit

```

using System;
using System.Data;
using System.Data.OleDb;
using System.Windows.Forms;

namespace StudentDatabaseApp
{
    public partial class Form1 : Form----
    {
        OleDbConnection con = new OleDbConnection(@"Provider=Microsoft.Jet.OLEDB.4.0;Data
Source=C:\std.mdb");
        DataSet d1 = new DataSet("student");
        int counter = 0;

        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
        {
            try
            {
                con.Open();
                OleDbDataAdapter adap = new OleDbDataAdapter("select * from student", con);
                adap.Fill(d1, "student");
                DisplayCurrentRecord();
            }
            catch (Exception ex)
            {
                MessageBox.Show("Error: " + ex.Message);
            }
        }

        private void DisplayCurrentRecord()
        {
            if (d1.Tables[0].Rows.Count > 0)
            {
                textBox1.Text = d1.Tables[0].Rows[counter]["ID"].ToString();
                textBox2.Text = d1.Tables[0].Rows[counter]["Name"].ToString();
                textBox3.Text = d1.Tables[0].Rows[counter]["Phone"].ToString();
                textBox4.Text = d1.Tables[0].Rows[counter]["Address"].ToString();
                textBox5.Text = d1.Tables[0].Rows[counter]["deptid"].ToString();
            }
        }

        private void button1_Click(object sender, EventArgs e) // First
        {
            counter = 0;
            DisplayCurrentRecord();
        }
    }
}

```

```

private void button1_Click(object sender, EventArgs e) // First
{
    counter = 0;
    DisplayCurrentRecord();
}

private void button2_Click(object sender, EventArgs e) // Previous
{
    if (counter > 0)
    {
        counter--;
        DisplayCurrentRecord();
    }
    else
    {
        MessageBox.Show("You are already on the first record");
    }
}

private void button3_Click(object sender, EventArgs e) // Next
{
    if (counter < d1.Tables[0].Rows.Count - 1)
    {
        counter++;
        DisplayCurrentRecord();
    }
    else
    {
        MessageBox.Show("You are already on the last record");
    }
}

private void button4_Click(object sender, EventArgs e) // Last
{
    counter = d1.Tables[0].Rows.Count - 1;
    DisplayCurrentRecord();
}

private void button5_Click(object sender, EventArgs e) // Display
{
    DisplayCurrentRecord();
}

private void button6_Click(object sender, EventArgs e) // Insert
{
    try
    {
        OleDbCommand com1 = new OleDbCommand(
            "INSERT INTO student(ID, Name, Phone, Address, deptid) " +
            "VALUES(' + textBox1.Text + "','" + textBox2.Text + "','" +
            textBox3.Text + "','" + textBox4.Text + "','" + textBox5.Text + "',')",
            con);
        com1.ExecuteNonQuery();
        MessageBox.Show("Record added successfully");

        // Refresh the dataset
        d1.Clear();
        OleDbDataAdapter adap = new OleDbDataAdapter("select * from student", con);
        adap.Fill(d1, "student");
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}

```



```

private void button7_Click(object sender, EventArgs e) // Update
{
    try
    {
        OleDbCommand com3 = new OleDbCommand(
            "UPDATE student SET Name='" + textBox2.Text + "', Phone='" + textBox3.Text +
            "', Address='" + textBox4.Text + "', deptid='" + textBox5.Text +
            "' WHERE ID=" + textBox1.Text, con);
        com3.ExecuteNonQuery();
        MessageBox.Show("Record updated successfully");

        // Refresh the dataset
        d1.Clear();
        OleDbDataAdapter adap = new OleDbDataAdapter("select * from student", con);
        adap.Fill(d1, "student");
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}

private void button8_Click(object sender, EventArgs e) // Delete
{
    try
    {
        OleDbCommand com2 = new OleDbCommand(
            "DELETE FROM student WHERE ID = " + textBox1.Text, con);
        com2.ExecuteNonQuery();
        MessageBox.Show("Record deleted successfully");

        // Refresh the dataset
        d1.Clear();
        OleDbDataAdapter adap = new OleDbDataAdapter("select * from student", con);
        adap.Fill(d1, "student");

        // Reset to first record
        counter = 0;
        DisplayCurrentRecord();
    }
    catch (Exception ex)
    {
        MessageBox.Show("Error: " + ex.Message);
    }
}

private void button9_Click(object sender, EventArgs e) // Join
{
    try
    {
        OleDbDataAdapter adap1 = new OleDbDataAdapter(
            "SELECT student.ID, dept.deptname FROM student " +
            "INNER JOIN dept ON dept.deptid=student.deptid", con);
        DataSet d2 = new DataSet();
        adap1.Fill(d2, "student_dept");

        // Create a new form to display the joined data
        Form joinForm = new Form();
        DataGridView dgv = new DataGridView();
        dgv.Dock = DockStyle.Fill;
        dgv.DataSource = d2.Tables["student_dept"];
        joinForm.Controls.Add(dgv);
        joinForm.Text = "Student-Department Join";
        joinForm.Show();
    }
}

```

```
    }  
    catch (Exception ex)  
    {  
        MessageBox.Show("Error: " + ex.Message);  
    }  
}  
}
```